A Mini Project Report on

Planora- A Blend of Architecture and Vision

Submitted in partial fulfillment of the requirements for the degree

Third Year Engineering – Computer Science and Engineering (Data Science)

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Academic year: 2025-26

CERTIFICATE

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ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Ms. Harsha Zope**. Expressing gratitude towards our HoD, **Dr. Pravin Adivarekar**, and the Department of Computer Science and Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Ms. Shubhangi Soni** who gave us her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

ABSTRACT

The project *Planora* is an intelligent digital platform designed to simplify the transformation of 2D floor plans into realistic and interactive 3D models, enhancing architectural visualization and user experience. The system supports multiple input formats such as JPG, PNG, PDF, and SVG, ensuring flexibility and accessibility for users. Advanced image preprocessing techniques using computer vision are applied to clean and structure the input data, enabling accurate detection of architectural elements. An AI-powered approach utilizing Region-Based Convolutional Neural Networks (Mask RCNN) is implemented to identify structural components such as walls, doors, and rooms from floor plans. The extracted features are then converted into spatial coordinates, which are used to generate detailed 3D models. The platform integrates powerful web-based visualization using Three.js, allowing users to explore the generated models through multiple viewing modes, including bird's-eye and walkthrough perspectives. The frontend is developed using React.js for an interactive and user-friendly interface, while the backend is powered by Python with Flask/FastAPI to handle processing and data flow efficiently. TensorFlow is utilized for model development and training, ensuring accurate and scalable AI performance. By combining artificial intelligence, computer vision, and modern web technologies, Planora provides a seamless solution for architects, designers, and users to visualize spaces efficiently, reducing manual effort and improving design understanding.

Keywords: *Planora, 2D to 3D conversion, Mask RCNN, computer vision, TensorFlow, Three.js, React.js, floor plan analysis, architectural visualization, AI-based modelling.*

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Chapter 1

Introduction

The process of visualizing architectural designs from 2D floor plans remains a significant challenge for users, particularly for students, designers, and clients who may lack technical expertise in interpreting technical drawings. Traditional floor plans often require professional knowledge to understand spatial relationships, dimensions, and structural elements, making it difficult for non-experts to accurately imagine the final outcome. This gap between 2D representation and real-world visualization can lead to misunderstandings, design inefficiencies, and increased dependency on manual modelling tools.

Existing solutions for 3D modelling typically involve complex software that demands advanced skills, significant time investment, and high computational resources. Many of these tools are not user-friendly and may not support direct conversion from 2D floor plans, requiring users to manually recreate designs in a 3D environment. This not only slows down the workflow but also increases the chances of errors and inconsistencies between the original plan and the generated model. As a result, there is a growing need for an automated and intelligent system that simplifies this conversion process while maintaining accuracy and usability.

The proposed system, *Planora*, addresses these challenges by enabling users to upload 2D floor plans in multiple formats such as JPG, PNG, PDF, and SVG, and automatically converting them into detailed 3D models using artificial intelligence. The system integrates computer vision techniques for preprocessing and feature extraction, along with an AI-based Region-Based Convolutional Neural Network (RCNN) to detect structural elements such as walls, doors, and rooms. These detected components are transformed into spatial coordinates, forming the basis for accurate 3D reconstruction.

To enhance usability and interaction, the platform provides an intuitive web interface built with React.js, allowing users to easily upload files and explore generated models. The 3D visualization is powered by Three.js, enabling real-time rendering and interactive viewing modes, including bird's-eye and walkthrough perspectives. The backend, developed using Python with Flask/FastAPI, ensures efficient processing and communication between system components, while TensorFlow supports the development and training of machine learning models.

Beyond improving visualization and design understanding, Planora contributes to efficiency and accessibility in architectural workflows. By automating the conversion process, it reduces manual effort, minimizes errors, and saves time for users. The platform makes advanced visualization tools more accessible to a wider audience, including students and non-professionals, thereby bridging the gap between technical design and practical understanding. Ultimately, Planora aims to transform how users interact with floor plans by providing a smarter, faster, and more intuitive approach to architectural visualization.

1.1 Purpose:

The purpose of this project is to develop an intelligent platform that simplifies the visualization of architectural designs by converting 2D floor plans into realistic and interactive 3D models. Many users, including students and non-professionals, find it difficult to interpret 2D layouts due to limited spatial understanding and lack of technical expertise. Additionally, existing 3D modeling tools are often complex and time-consuming, creating a need for a more accessible and automated solution. This project aims to bridge that gap by providing an easy-to-use system that transforms 2D plans into accurate 3D representations with minimal effort.

The system leverages advanced technologies such as computer vision and AI-based models, including RCNN, to detect structural elements like walls, doors, and rooms from uploaded floor plans. These features are processed and converted into spatial data to generate detailed 3D models. The platform supports multiple file formats such as JPG, PNG, PDF, and SVG, ensuring flexibility for users. Additionally, interactive 3D visualization powered by Three.js allows users to explore the generated models through different viewing modes, enhancing overall usability and understanding.

Ultimately, the project is designed to make architectural visualization more efficient, accessible, and user-friendly. By automating the conversion process and reducing manual effort, it helps users save time and minimize errors while improving design clarity. The platform aims to support better decision-making, enhance learning experiences, and provide a modern approach to exploring and understanding spatial layouts.

1.2 Problem Statement:

The problem statement for *Planora* revolves around the challenges users face in understanding and visualizing architectural designs from 2D floor plans. Traditional floor plans require technical expertise to interpret spatial layouts, dimensions, and structural elements, making them difficult for students, clients, and non-professionals to fully comprehend. Existing 3D modeling solutions are often complex, time-consuming, and require manual effort to recreate designs, which increases the chances of errors and inefficiencies. Additionally, many current systems lack automation, flexibility in file formats, and interactive visualization features, limiting their usability and accessibility. Planora aims to address these challenges by providing an AI-powered platform that automatically converts 2D floor plans into accurate and interactive 3D models, improving design understanding and efficiency.

1. Difficulty in Interpreting 2D Floor Plans: Users often struggle to understand spatial relationships and layouts from 2D drawings, especially without technical or architectural knowledge.
2. Manual and Time-Consuming 3D Modeling: Existing tools require users to manually recreate floor plans in 3D environments, which is time-intensive and prone to human error.
3. Lack of Automation in Conversion: Many systems do not provide automated 2D-to-3D conversion using AI, making the process inefficient and dependent on skilled professionals.
4. Limited Support for Multiple File Formats: Current solutions often do not support diverse input formats like JPG, PNG, PDF, and SVG, reducing flexibility for users.
5. Absence of Interactive Visualization: Traditional systems lack advanced visualization features such as walkthroughs, bird's-eye views, and customizable rendering, limiting user experience and design clarity.

1.3 Objectives:

In a bid to transform the way architectural designs are visualized and understood, Planora sets out with clear objectives. It aims to simplify the conversion of 2D floor plans into interactive 3D models while improving accessibility and efficiency in design workflows. By leveraging deep learning techniques, automated model generation, and real-time visualization, Planora seeks to create a unified platform where both technical and non-technical users can easily explore and understand spatial layouts. Through innovation and continuous improvement, Planora aspires to enhance design accuracy, usability, and modern architectural practices.

1. To develop an AI-based detection system for accurate identification and segmentation of architectural elements such as walls, doors, and rooms from 2D floor plans using Mask R-CNN.
2. To enhance layout accuracy, ensure structural consistency, and optimize space utilization in floor plans by applying rule-based algorithms and spatial optimization techniques.
3. To enable users to upload 2D floor plans in formats such as JPG, PNG, PDF, and SVG and automatically convert them into precise and interactive 3D models using AI-based methods.
4. To provide customizable and interactive 3D visualization, including lighting, textures, material settings, and multiple viewing modes such as bird's-eye view and walkthrough navigation using Three.js.

1.4 Scope:

The scope of the Planora project is to develop a user-friendly platform that automatically converts 2D floor plans into accurate and interactive 3D models. By integrating deep learning techniques for object detection and real-time visualization, the system enhances architectural understanding and accessibility. It also simplifies the design process by reducing manual effort and eliminating the need for complex CAD tools, making it suitable for both technical and non-technical users.

1. Provide an Automated Conversion System: Develop a system that can efficiently convert 2D floor plans into 3D models, creating a streamlined and reliable workflow for architectural visualization.

2. Ensure Accessibility and Ease of Use: Design a platform that is simple and intuitive, allowing users without technical or CAD expertise to easily visualize and explore building layouts.

3. Enhance Visualization and User Experience: Offer interactive 3D viewing options such as bird's-eye view and walkthrough mode, allowing users to better understand spatial layouts and

design structures.

4. Incorporate AI and Computer Vision Technologies: Integrate RCNN for object detection and OpenCV for image preprocessing to accurately identify structural elements like walls, doors, and rooms, improving model precision.

Chapter 2

Literature Review

The literature review explores the advancements in floor plan analysis and 3D reconstruction techniques, which are essential for improving architectural visualization and automation. With the increasing demand for smart design systems, researchers have focused on integrating computer vision, deep learning, and 3D modelling techniques. This includes object detection, image segmentation, and spatial reconstruction methods that enable accurate identification of structural components such as walls, doors, and windows. The review aims to analyze key methodologies, challenges, and recent developments while identifying gaps for further improvement in automated floor plan interpretation systems.

N. Bróz, P. Nazarko, A. Prokop, and A. Borowiec (2025) [1] proposed a deep learning-based approach for recognizing building elements in residential floor plan drawings. Their study focuses on detecting structural components such as walls, doors, and windows using advanced object detection and segmentation models. The methodology involves training deep neural networks on annotated floor plan datasets to improve detection accuracy. The results demonstrate significant improvement in identifying architectural elements, highlighting the effectiveness of deep learning techniques in automating floor plan analysis.

R. Khade, K. Jariwala, and C. Chattopadhyay (2025) [2] conducted a comprehensive survey of floor plan image analysis and related applications. Their research reviews various AI-based techniques including deep learning, object detection, and spatial analysis methods used for extracting layout information. The study identifies key challenges such as data variability, complex layouts, and accuracy limitations. The findings suggest that deep learning models play a crucial role in improving automated floor plan analysis and layout extraction, making them suitable for real-world applications.

D. Al-Turki et al. (2023) [3] introduced a progressive active learning approach for floor plan image analysis. Their method focuses on improving detection performance by using annotated datasets and iterative learning techniques. The approach reduces manual annotation effort while enhancing model accuracy. The results show better recognition of floor plan components and improved efficiency in model training, demonstrating the potential of active learning in optimizing AI-based architectural systems.

Chapter 3

Proposed System

Planora is an AI-driven platform designed to transform 2D floor plans into interactive 3D models. By leveraging computer vision techniques and Region-based Convolutional Neural Networks (RCNN), the system identifies key architectural components such as walls, doors, and rooms from uploaded floor plan images. It supports multiple input formats, including JPG, PNG, PDF, and SVG, ensuring flexibility for users.

The system utilizes OpenCV for image preprocessing and TensorFlow for model training and inference, enabling accurate feature extraction and reconstruction. The generated 3D models are rendered using Three.js, allowing users to interact with and explore designs in real time. Overall, Planora offers a fast, automated, and user-friendly solution for architectural visualization.

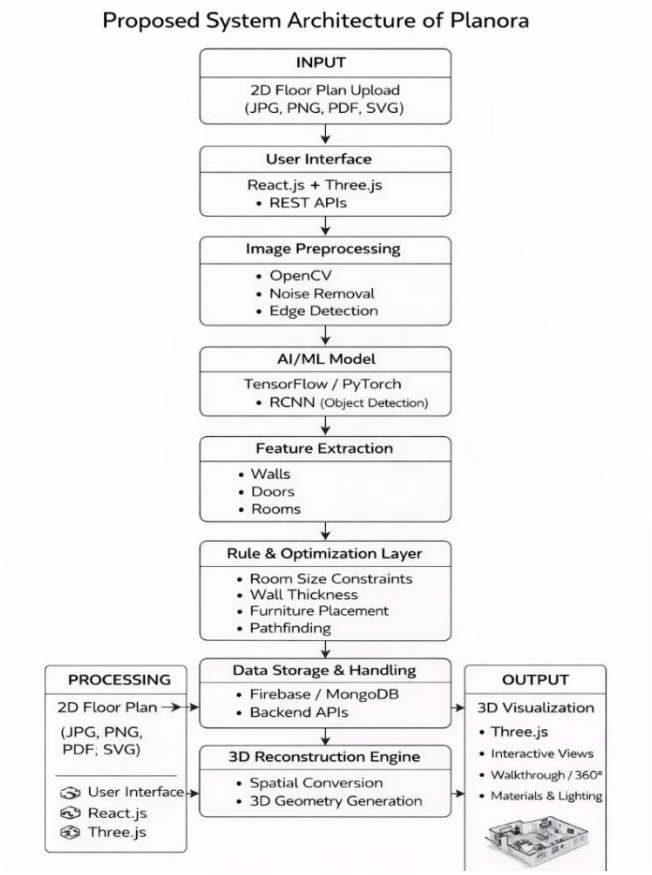

Fig 3.1 Proposed System Architecture of Planora

Once the floor plan is uploaded, it is handled by the backend processing system, implemented using Python or Node.js. In this stage, OpenCV is used for image preprocessing tasks such as noise reduction and edge detection. The processed image is then passed to an AI/ML model built with TensorFlow and RCNN, which detects structural elements such as walls, doors, and rooms. This ensures accurate feature extraction and reliable interpretation of the floor plan.

The extracted data is then managed through backend APIs and stored in databases such as Firebase or MongoDB. This data handling and storage layer ensures efficient storage, retrieval, and communication between different system components, enabling smooth and scalable system performance.

Following this, the system performs 3D reconstruction and visualization. The extracted structural information is converted into spatial coordinates to generate a 3D model. This model is rendered using Three.js, allowing users to interact with it through various viewing modes, including bird's-eye view and walkthrough. Additionally, users can customize lighting, textures, and materials to achieve realistic visualization. The interface, supported by React.js and a secure backend such as Flask or FastAPI, ensures smooth performance and an intuitive user experience.

3.1 Features and Functionality:

The Planora platform integrates artificial intelligence, computer vision, and modern web technologies to provide an efficient and user-friendly solution for architectural visualization. One of its key features is automated 2D-to-3D conversion, where users can upload floor plans in formats such as JPG, PNG, PDF, and SVG, and the system automatically converts them into accurate 3D models. This eliminates the need for manual modeling, significantly reducing time and effort.

Another important feature is AI-based structural detection. The system utilizes RCNN and TensorFlow, combined with OpenCV preprocessing, to identify architectural elements such as walls, doors, and rooms. This ensures precise feature extraction and enhances the reliability of the generated models.

The platform also provides interactive 3D visualization, where the generated models are rendered using Three.js. Users can explore designs through multiple perspectives, including bird's-eye views and walkthrough modes, improving spatial understanding and decision-making.

Chapter 4

Requirement and Analysis

For the requirement analysis of the Planora project, we need to identify and document the functional and non-functional requirements that the system must meet to fulfill its objectives effectively. Here's a breakdown of the requirement analysis for Planora:

A. Functional Requirements:

1. User Input and Upload Files: The system should allow users to upload 2D floor plans in formats such as JPG, PNG, PDF, and SVG. The upload process should be simple and support multiple file types for flexibility.

2. Image Preprocessing: The system should preprocess uploaded images using computer vision techniques such as noise removal, resizing, and edge detection (OpenCV) to improve image quality for further processing.

3. AI-Based Element Detection: The system should implement deep learning models such as Mask R-CNN/TensorFlow to detect and segment architectural elements like walls, doors, windows, and rooms from floor plans.

4. 2D to 3D Model Generation: The system should convert detected elements into spatial coordinates and automatically generate an accurate 3D model that represents the original layout.

5. 3D Visualization and Interaction: Users should be able to view and interact with the generated 3D model using features such as rotation, zoom, bird’s-eye view, and walkthrough mode using Three.js.

6. Customization Feature: The system should allow users to customize the 3D model by adjusting materials, textures, lighting, and rendering styles to enhance realism.

7. Save and Download Functionality: Users should be able to save or download the generated 3D models for future use and sharing.

B. Non-Functional Requirements:

1. Performance: The system should ensure fast processing of floor plans, including preprocessing, detection, and 3D generation. The response time should be minimal to provide a smooth user experience even with multiple users.

2. Scalability: The system should be scalable to handle increasing numbers of users, larger datasets, and more complex floor plans. It should support cloud-based deployment for better scalability.

3. Usability: The interface should be user-friendly, intuitive, and responsive. Even non-technical users should be able to upload floor plans and interact with 3D models easily without confusion.

4. Maintainability: The system should have a modular and well-structured codebase, allowing easy updates, bug fixes, and integration of new features such as improved AI models or additional visualization tools.

5. User Experience: The platform should provide a smooth and engaging experience with fast loading times, interactive 3D visualization, and easy navigation. Features like walkthrough and customization should enhance user satisfaction.

Chapter 5

Project Design

5.1 Use Case Diagram:

This Use Case Diagram visually represents the functions and interactions within your Online Bookstore System. It clearly defines who interacts with the system (Actors) and what they can do (Use Cases), along with the automated processes that support those actions.

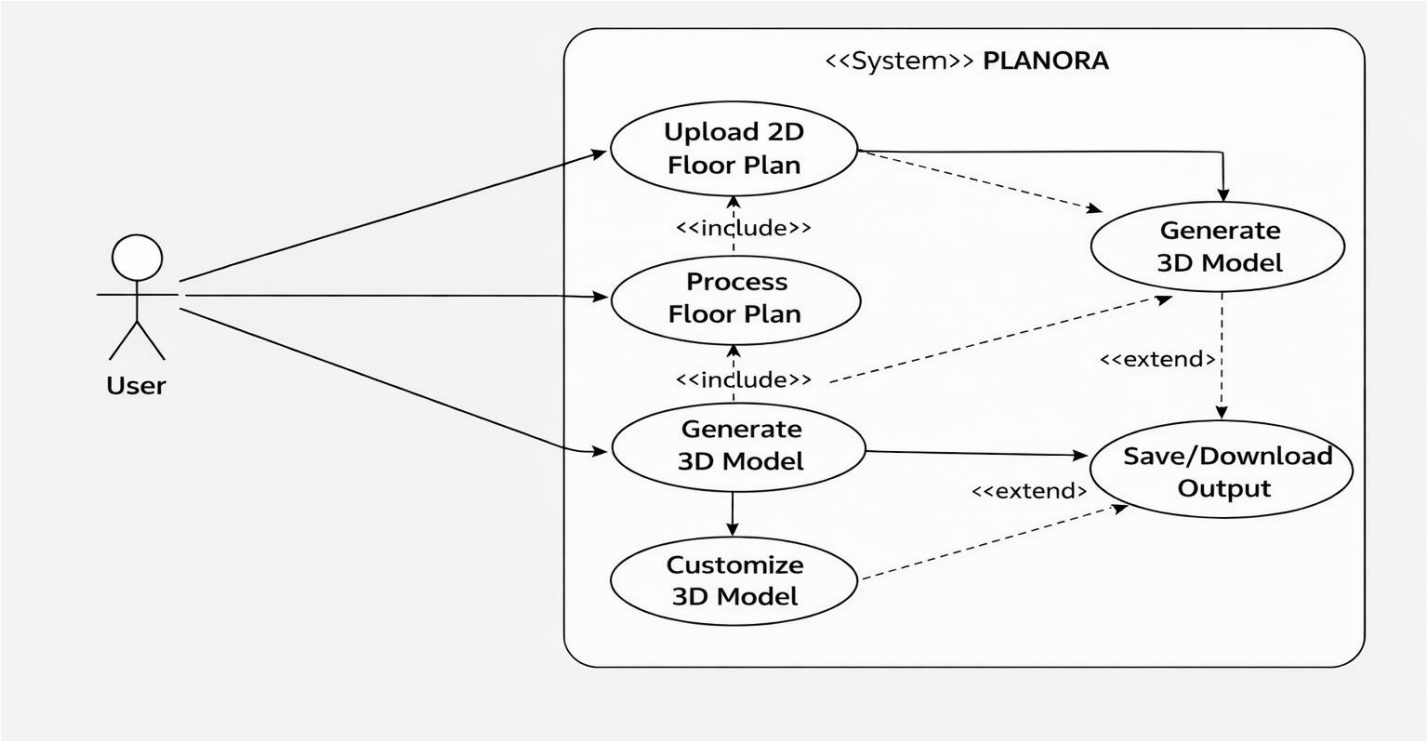

Fig 5.1 Illustrates the Use case Diagram of the system.

The Use Case Diagram of PLANORA provides a comprehensive view of how a user interacts with the system to transform a 2D floor plan into a fully visualized 3D model. In this system, the user acts as the primary and only actor, initiating and controlling all operations through an interactive interface. The process begins when the user uploads a 2D floor plan in supported formats such as JPG, PNG, PDF, or SVG. This step serves as the entry point of the system, where the user provides the essential input required for further processing.

Once the floor plan is uploaded, the system performs image preprocessing using computer vision techniques. This includes operations such as noise removal, resizing, and edge detection to enhance the clarity and structure of the image. The goal of this stage is to ensure that the input data is clean and suitable for accurate analysis by the AI model. After preprocessing, the system proceeds to the element detection phase, which is the core functionality of the application. Here, advanced deep learning algorithms like Mask R-CNN are used to identify and classify various architectural components present in the floor plan, such as walls, doors, windows, and furniture. This step is crucial as it extracts meaningful information from the image and forms the basis for 3D reconstruction.

Following the detection phase, the system performs the 2D to 3D conversion. In this stage, the identified elements are transformed into a three-dimensional structure by assigning spatial coordinates, dimensions, and relationships between objects. The system constructs a digital 3D model that accurately represents the layout of the original floor plan. Once the 3D model is generated, the user is given the ability to customize the design. This includes modifying materials (such as wood, glass, or tiles), adjusting lighting conditions, and selecting different rendering styles. These customization features enhance the realism and usability of the model, allowing users to tailor the output according to their preferences.

After customization, the user can view the 3D model using multiple visualization modes. These include a bird's-eye view, which provides an overall layout of the structure, and a walkthrough mode, which allows the user to virtually navigate inside the building. This interactive visualization significantly improves the understanding of spatial arrangements and design details. Finally, the system provides an option to download or save the generated 3D model, enabling the user to store the output for future use, sharing, or further modifications.

In summary, the use case diagram captures the entire workflow of the PLANORA system, illustrating how the user interacts with various functionalities from input to output. It emphasizes the sequence of operations and the user's role in controlling the process, while abstracting away the internal technical complexities.

5.2 DFD (Data Flow Diagram):

This DFD outlines the core functionalities of the Planora platform from user interaction to system-level operations.

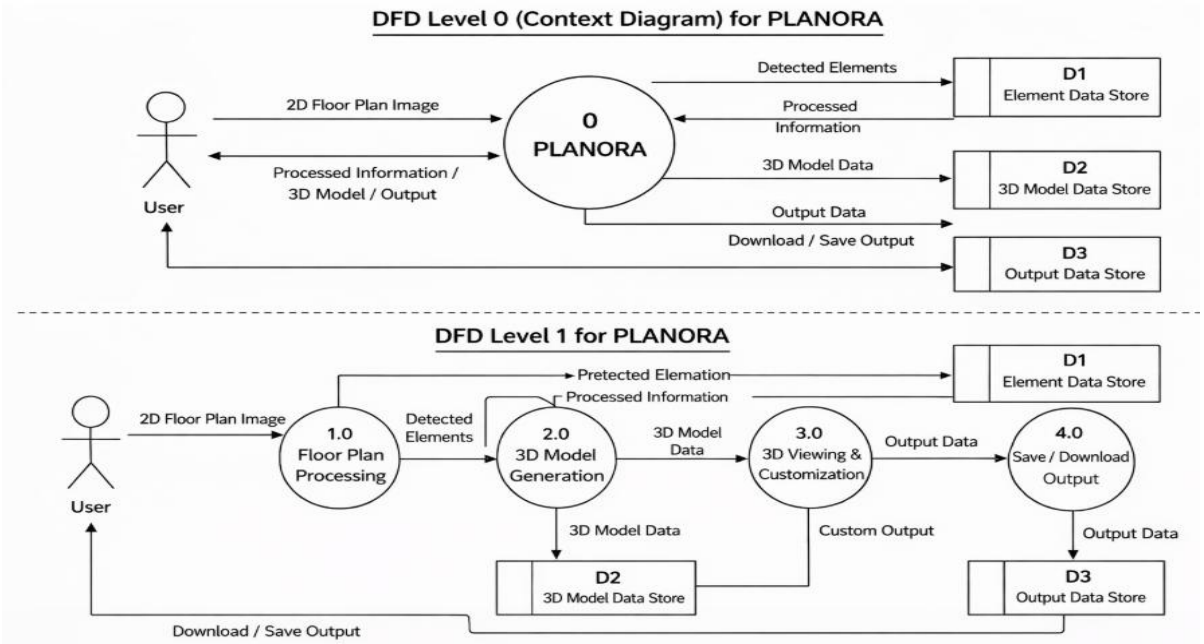

Fig 5.2 Data Flow Diagram

The Data Flow Diagram (DFD) of PLANORA illustrates how data moves through the system and how it is transformed from a 2D floor plan into a 3D model. It focuses on the internal processing of data rather than user interaction. The process begins with the user, who acts as the external entity and provides input to the system by uploading a 2D floor plan in formats such as JPG, PNG, PDF, or SVG. This input is then passed to the floor plan processing module, where the image undergoes preprocessing using computer vision techniques like noise removal, resizing, and edge detection. This step ensures that the image is clean and structured for further analysis.

After preprocessing, the data flows into the AI detection module, which is a crucial part of the system. In this module, deep learning algorithms such as Mask R-CNN are used to identify and extract important architectural elements like walls, doors, windows, and furniture. The detected components are then stored in an intermediate data store, often referred to as the element data store, which keeps structured information about the identified objects. This stored data is then passed to the 3D model generation module, where the system converts the 2D information into a

3D structure by assigning spatial coordinates, dimensions, and relationships between elements.

Once the 3D model is generated, it is stored in the 3D model data store and forwarded to the visualization and customization module. In this stage, technologies like Three.js are used to render the 3D model in a web environment. The user can interact with the model by applying different materials, adjusting lighting, and selecting various viewing modes such as bird's-eye view or walkthrough mode. The final processed output is then stored in the output data store, from where it can be accessed, viewed, or downloaded by the user.

Overall, the DFD clearly represents the step-by-step flow of data within the PLANORA system, starting from user input, moving through preprocessing, AI-based detection, and 3D generation, and ending with visualization and output. It highlights how different modules interact with each other and how data is stored and transformed at each stage, providing a clear understanding of the system's internal working.

5.3 System Architecture:

The system architecture of PLANORA follows a structured pipeline that converts a 2D floor plan into a 3D model through multiple stages, including input, processing, 3D generation, and output. The process begins with the user, who interacts with the system through a frontend interface developed using React.js. The user uploads a 2D floor plan in supported formats such as JPG, PNG, PDF, or SVG, which serves as the input to the system.

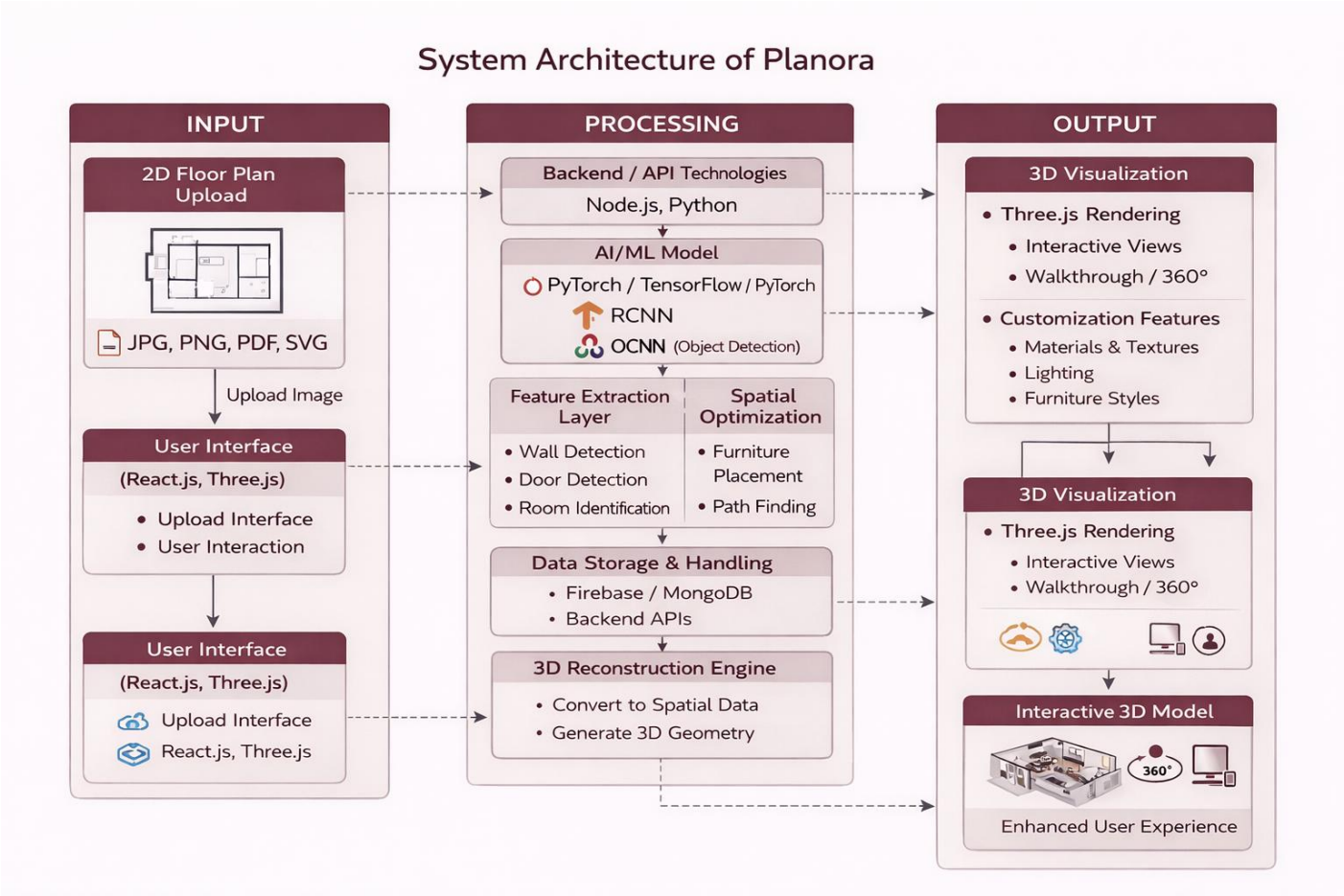

Fig 5.3 System Architecture

Once the input is received, it moves to the processing stage, where image preprocessing is performed using OpenCV. This includes operations such as noise reduction, edge detection, and image enhancement to ensure that the floor plan is clean and suitable for analysis. After preprocessing, the system uses an AI-based element detection module, implemented using deep learning models like Mask R-CNN in TensorFlow. This module identifies and classifies important

architectural components such as walls, doors, windows, and furniture from the floor plan.

The processed and detected data is then passed to the 3D generation stage, where the system constructs a three-dimensional model. In this stage, the detected elements are converted into spatial coordinates and mapped into a 3D structure using algorithms and rendering logic. The 3D model builder ensures that the layout, proportions, and relationships between elements are maintained accurately.

Following this, the system enters the output stage, where the generated 3D model is rendered using Three.js, a WebGL-based library. This allows users to interact with the model in real time through features such as rotation, zooming, bird's-eye view, and walkthrough mode. Additionally, the system provides customization options, enabling users to modify materials, textures, and lighting to enhance the visual appearance of the model.

Finally, the completed model can be saved or downloaded by the user for future use. The backend, built using Python frameworks like Flask or FastAPI, manages data processing, communication between modules, and integration of AI components. Overall, the system architecture ensures a smooth flow of data from input to output, combining frontend interaction, backend processing, AI intelligence, and 3D visualization into a unified and efficient system.

5.4 Implementation:

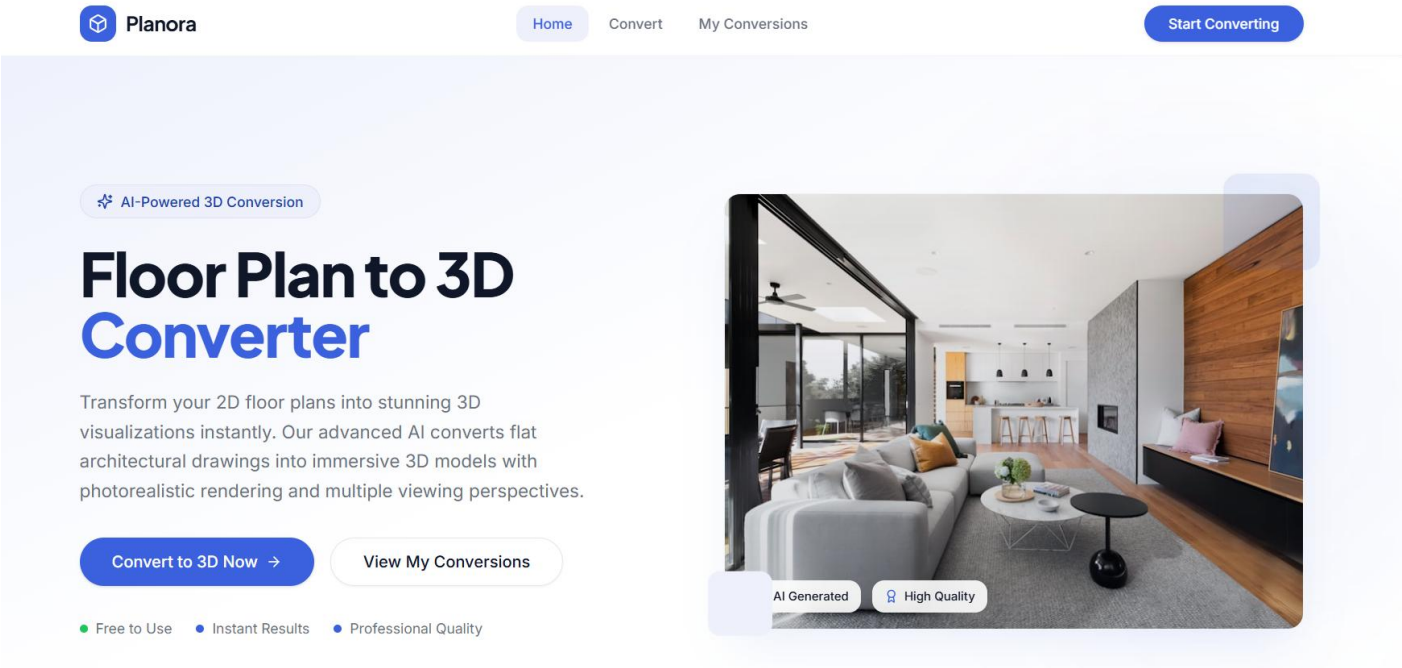

Fig 5.4 Homepage

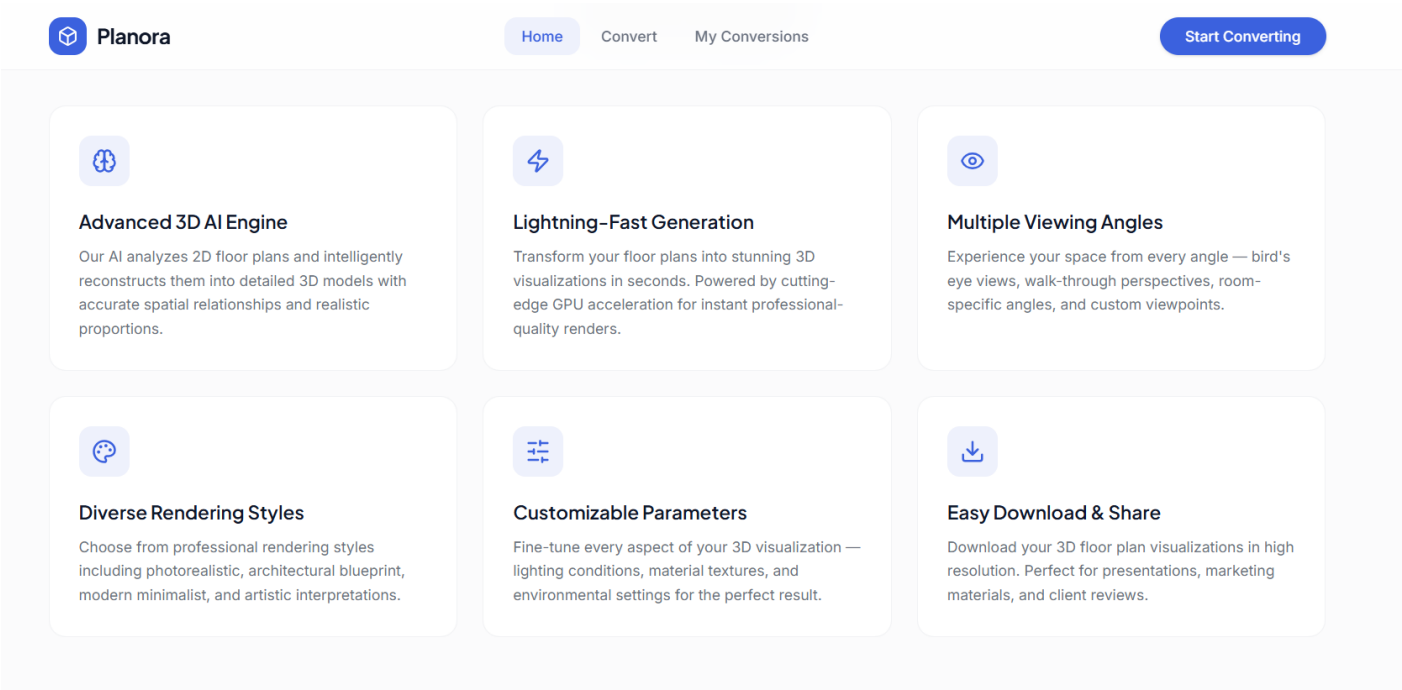

Fig 5.5 Features

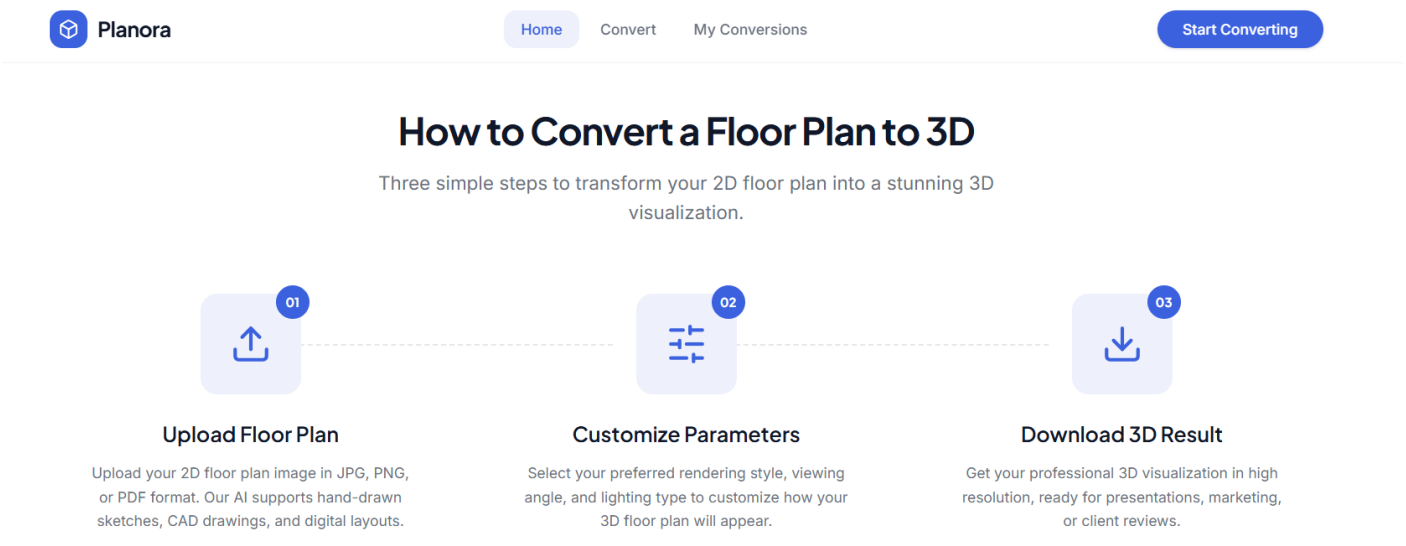

Fig 5.6 How to Convert Floor Plan to 3D

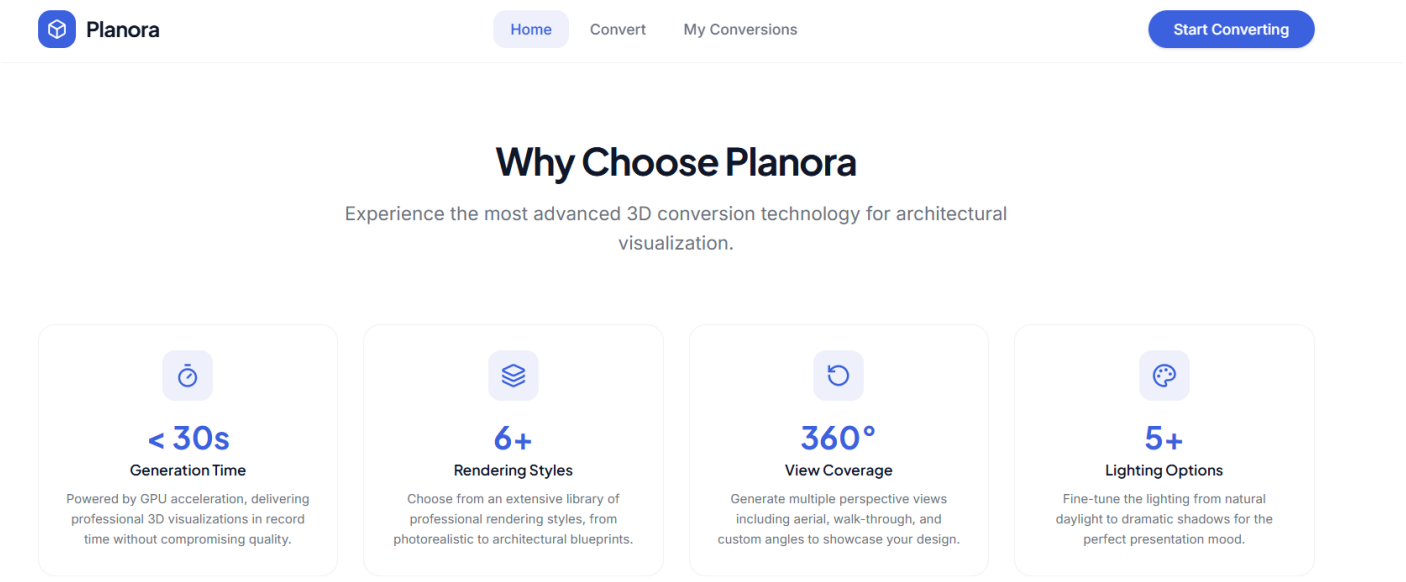

Fig 5.8 Why Choose Planora

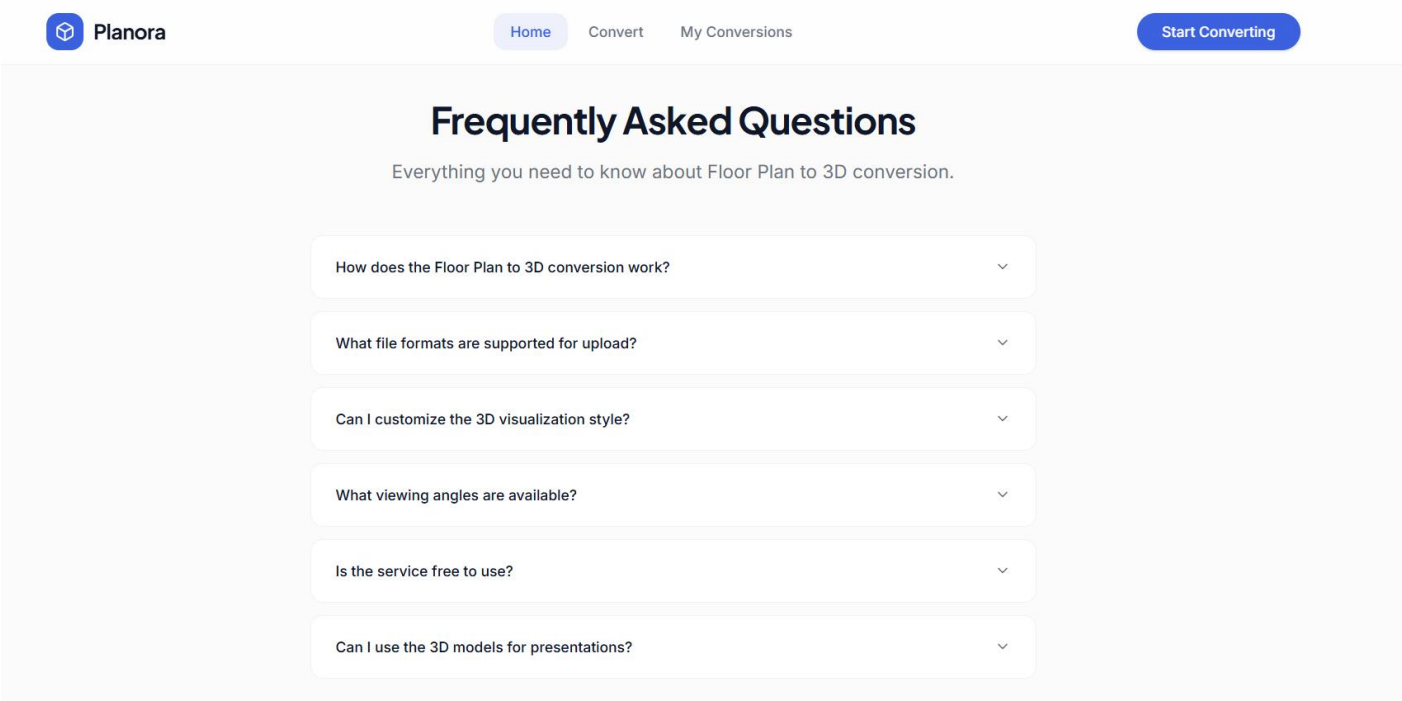

Fig 5.9 Frequently Asked Questions

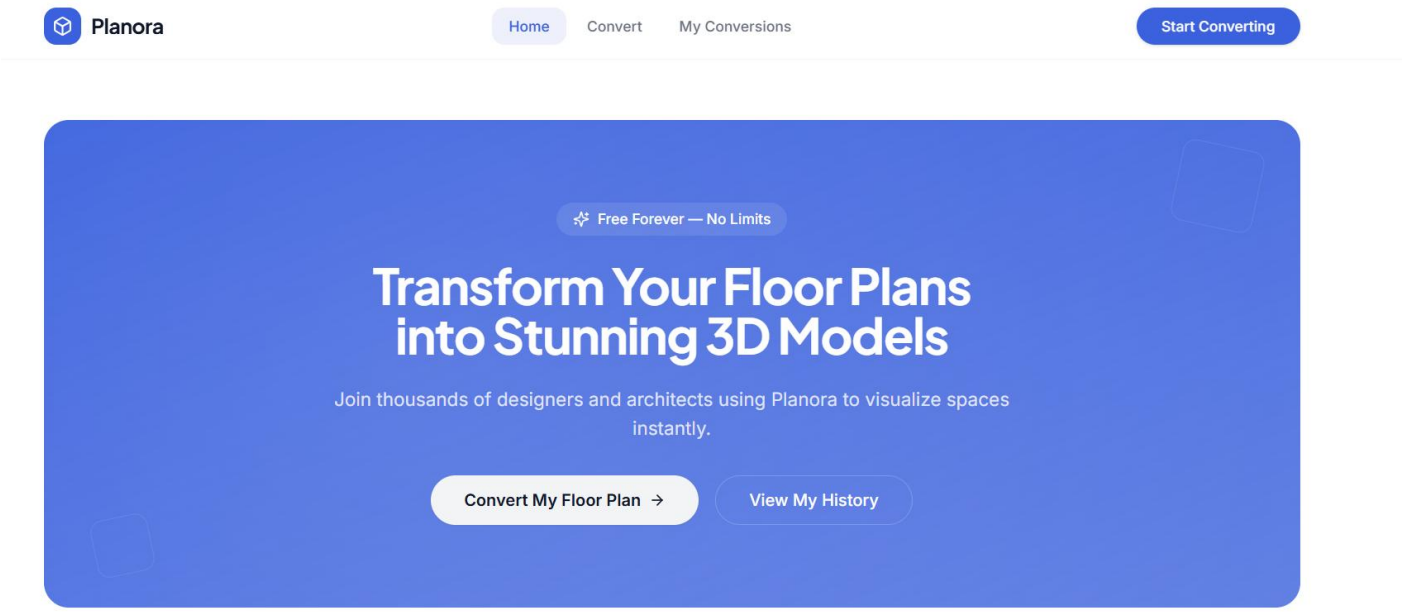

Fig 5.10 UI to convert 2D to 3

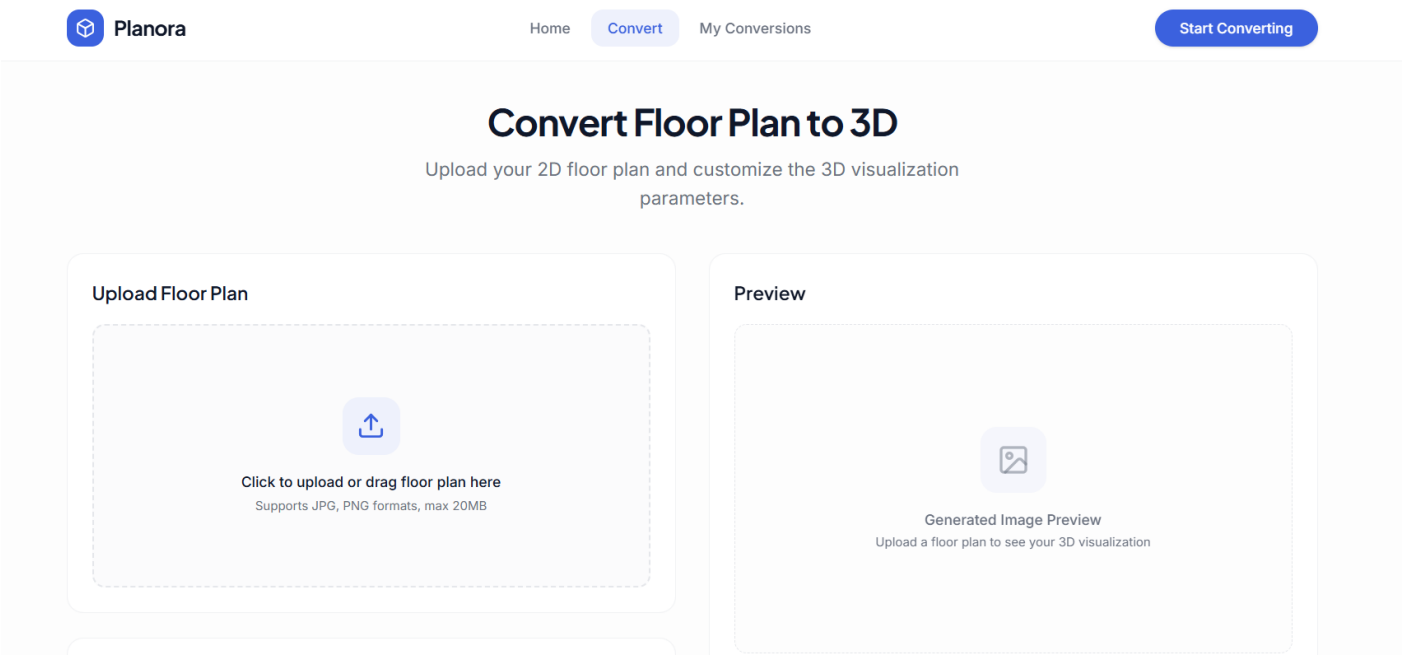

Fig 5.11 Convert Floor Plan to 3D

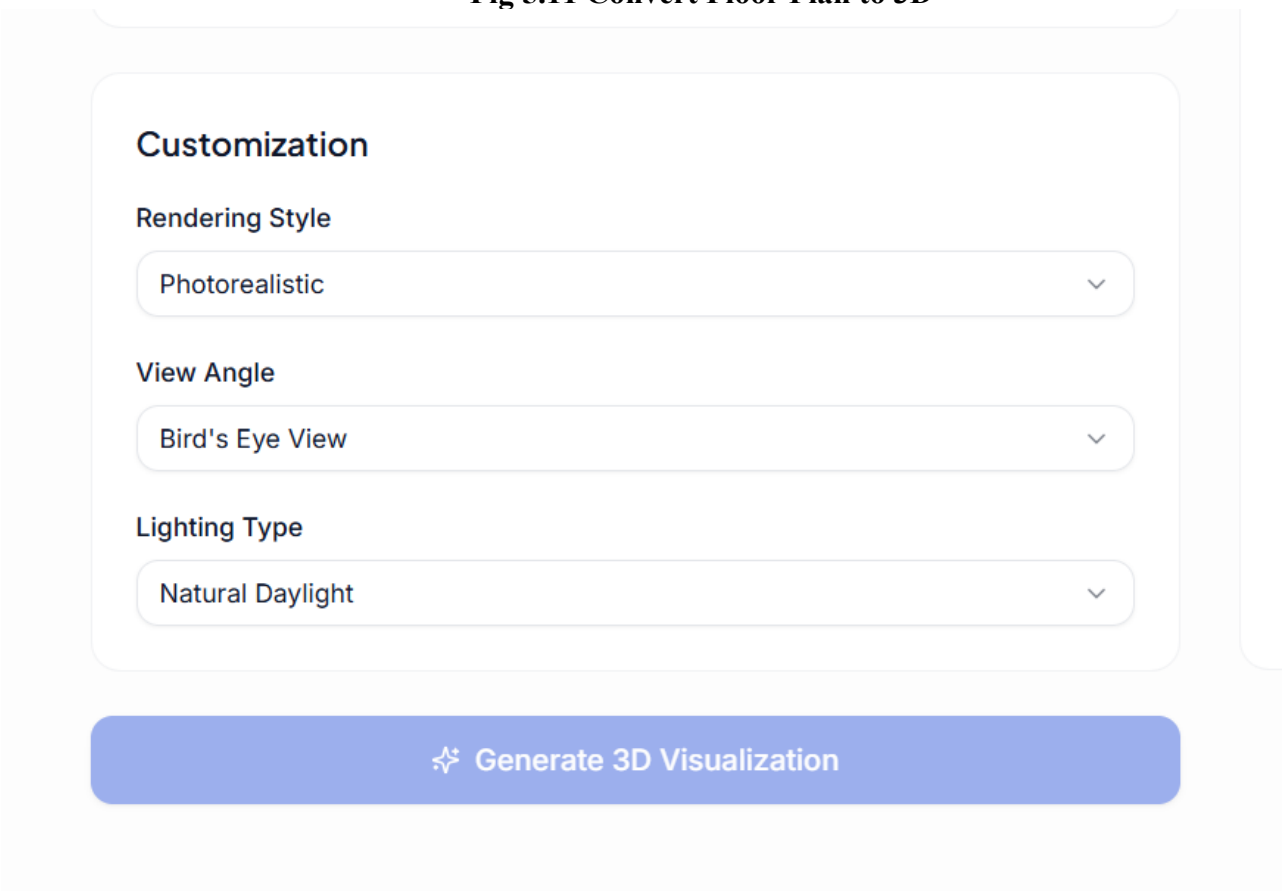

Fig 5.12 Customization Options

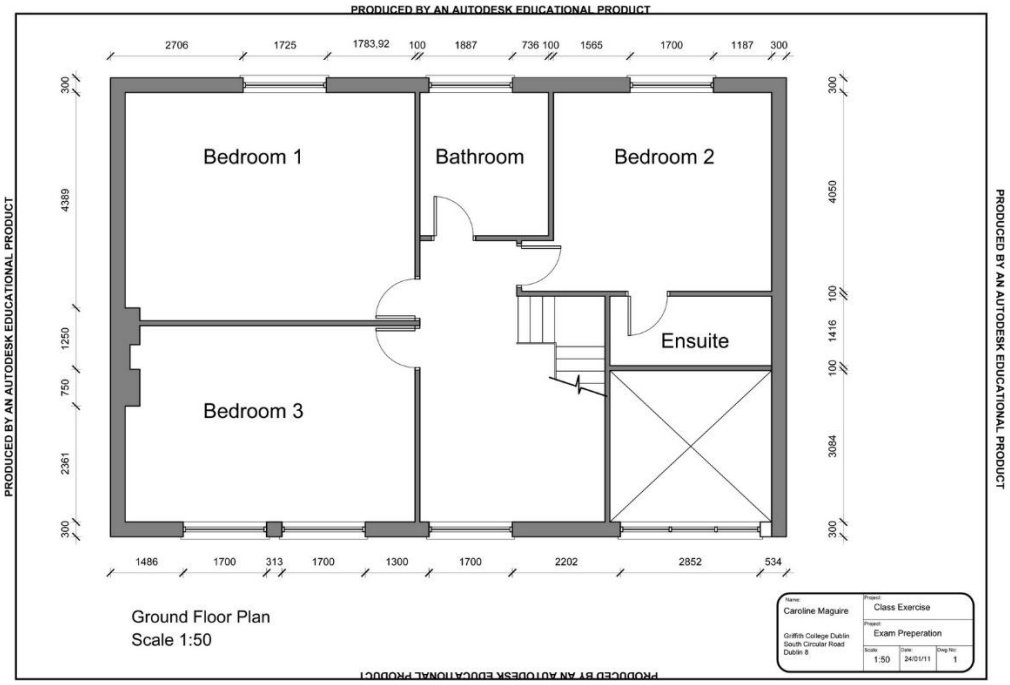

Fig 5.13 Implementation (2D Floor Plan)

Fig 5.13 Implementation (3D Floor Plan)

Chapter 6

Technical Specification

The project scheduling of PLANORA outlines the systematic planning and execution of different phases involved in the development of the system. It ensures that the project is completed within a defined timeline while maintaining efficiency and quality. The development process was divided into multiple phases, starting from requirement analysis to final documentation.

Initially, the project began with the requirement analysis phase, where the problem was identified and the objectives of converting 2D floor plans into 3D models were clearly defined. This phase involved understanding existing systems, identifying their limitations, and determining the scope of the proposed solution. Following this, a literature survey was conducted to study existing research papers and technologies related to floor plan analysis, deep learning, and 3D visualization.

After gathering sufficient knowledge, the project moved to the system design phase, where the architecture, use case diagrams, and data flow diagrams were created. This phase helped in planning how different components such as frontend, backend, and AI modules would interact with each other.

The next phase was the development phase, which involved implementing the frontend using React.js and the backend using Python frameworks like Flask or FastAPI. During this stage, image processing techniques using OpenCV and AI model integration using TensorFlow (Mask R-CNN) were also carried out. This phase required the most time and effort as it included coding, integration, and testing of various modules.

Once the development was completed, the project entered the testing and debugging phase, where the system was tested with different floor plan inputs to ensure accuracy and performance. Errors and bugs were identified and resolved to improve system reliability.

Finally, the project concluded with the documentation phase, where all the work, including design, implementation, and results, was compiled into a structured repository.

Chapter 7

Project Scheduling

Project scheduling for PLANORA was carried out in a structured and systematic manner to ensure timely completion of all phases. The entire development process was divided into two major phases: Phase 1 (Planning and Design) and Phase 2 (Development and Implementation). Each phase consists of multiple activities distributed across a period of twelve weeks.

Sr. No	Group members	Duration	Task Performed
1.	Simarpreet Kaur, Gauri Kandarkar, Riya More, Adarsh Panigrahi	2nd week of January	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project.
2.	Simarpreet Kaur, Gauri Kandarkar, Riya More	3rd week of January	Identifying the functionalities of the Mini Project.
3.	Riya More, Adarsh Panigrahi	1st week of February	Discussing the project topic with the help of paper prototype.
4.	Simarpreet Kaur, Gauri Kandarkar, Riya More	2nd week of February	Designing the Graphical User Interface (GUI)
5.	Simarpreet Kaur, Gauri Kandarkar	1st week of March	Adding the features of the Planora Website.
6.	Simarpreet Kaur, Gauri Kandarkar, Riya More	3rd week of March	Database design and connectivity
7.	Simarpreet Kaur, Gauri Kandarkar, Riya More, Adarsh Panigrahi	1st week of April	Integrating the model on GUI and connecting with the database.

Table 7.1: Project Task Distribution

A Gantt chart is a type of bar chart that illustrates a project schedule. This chart lists the tasks to be performed on the vertical axis, and time intervals on the horizontal axis. Gantt chart (Fig 7.1) illustrates the start and finish dates of the terminal elements and summary elements of a project.

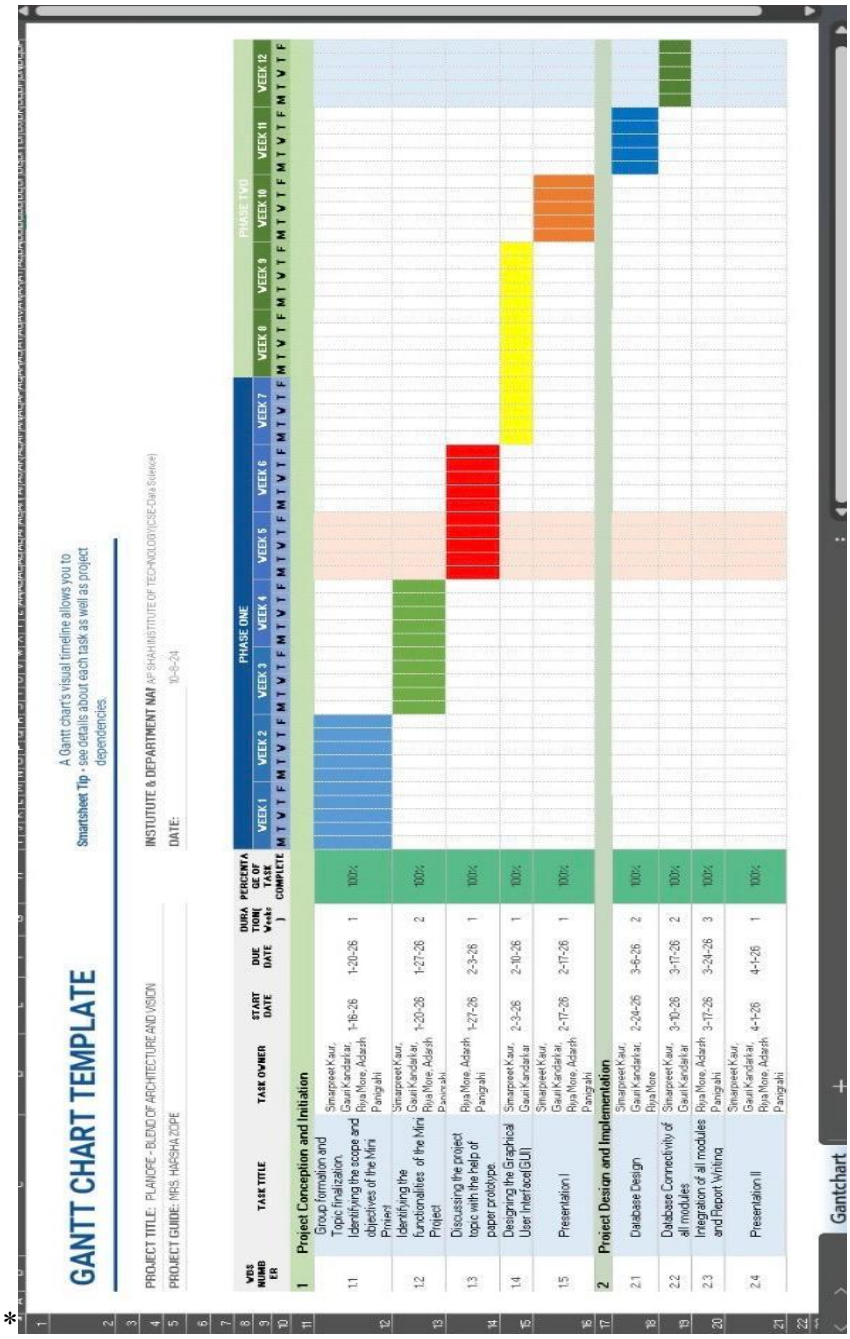

Fig 7.2 Gantt Chart of Read-Rover

Chapter 8

Results

The PLANORA system was successfully developed and evaluated for its ability to convert 2D floor plans into accurate and interactive 3D models. The system was tested using a variety of floor plan inputs in different formats such as JPG, PNG, and PDF to ensure flexibility and robustness. The results indicate that the system performs effectively in identifying and extracting key architectural elements, including walls, doors, windows, and basic furniture components, using deep learning techniques such as Mask R-CNN.

During the testing phase, it was observed that the AI-based detection module achieved satisfactory accuracy in recognizing structural elements from standard floor plans. The preprocessing stage using OpenCV significantly improved image quality, which in turn enhanced detection performance. The system was able to correctly interpret spatial layouts and relationships between different components, forming a strong foundation for 3D reconstruction.

The 3D model generation module successfully transformed the extracted 2D information into a three-dimensional representation. The generated models preserved the overall layout, proportions, and positioning of elements, resulting in a realistic visualization of the building structure. The use of Three.js enabled efficient rendering of 3D models directly in the browser, providing smooth navigation and interaction. Users were able to view the models from multiple perspectives, including bird's-eye view and walkthrough mode, which improved the understanding of interior spaces and design flow.

Another significant outcome of the system is the customization capability, which allows users to modify materials, textures, and lighting conditions. This feature enhances the visual appeal of the generated models and makes the system more practical for real-world applications such as interior design and architectural planning. The user interface was found to be intuitive and easy to use, enabling users with minimal technical knowledge to operate the system effectively.

In terms of performance, the system demonstrated reasonable processing time for standard floor plans. The combination of optimized image processing and efficient AI model integration ensured that the workflow from input to output was smooth and relatively fast. However, it was observed that the processing time increases with the complexity and size of the floor plan.

Despite its strengths, the system has certain limitations in the results. In cases where the floor plans were highly complex, irregular, or of low image quality, the detection accuracy slightly decreased.

Additionally, the system relies on predefined datasets such as CubiCasa5K, which may not cover all possible variations of architectural designs. As a result, some elements may be misclassified or missed entirely. The generated 3D models, while accurate in structure, are still approximations and may not fully represent real-world dimensions or fine details.

Overall, the results demonstrate that the PLANORA system effectively achieves its objective of automating the conversion of 2D floor plans into 3D models. It provides a reliable, efficient, and user-friendly solution that enhances visualization and reduces manual effort. The system shows strong potential for future improvements and real-world deployment in fields such as architecture, construction, and interior design.

Chapter 9

Conclusion

The PLANORA system successfully demonstrates an efficient and intelligent approach for converting 2D floor plans into interactive 3D models using artificial intelligence and computer vision techniques. The project addresses the limitations of traditional methods by automating the process of architectural visualization, thereby reducing manual effort, time consumption, and dependency on technical expertise.

Through the integration of advanced technologies such as deep learning (Mask R-CNN), image processing (OpenCV), and 3D rendering (Three.js), the system is capable of accurately detecting structural elements like walls, doors, windows, and furniture, and transforming them into a coherent 3D representation. The frontend developed using React.js ensures a user-friendly interface, allowing users to easily upload floor plans, interact with the generated models, and customize various aspects such as materials and lighting.

One of the key strengths of the system lies in its ability to provide interactive visualization, including bird's-eye view and walkthrough modes, which significantly enhance the understanding of spatial layouts. Additionally, the customization features add flexibility and improve the overall user experience, making the system suitable for practical applications in architecture and interior design.

Although the system performs well for standard floor plans, certain limitations exist in handling highly complex or low-quality inputs. The accuracy of detection and 3D reconstruction depends on the quality of the dataset and the trained model. Despite these challenges, the system achieves a good balance between performance, accuracy, and usability.

In conclusion, PLANORA provides a scalable and effective solution for modern architectural visualization by leveraging AI-driven automation. It has the potential to be further enhanced with improved models, larger datasets, and advanced features, making it a valuable tool for future developments in the construction and design industry.

Chapter 10

Future Scope

The PLANORA system provides a solid foundation for automated 2D to 3D floor plan conversion; however, there are several opportunities for further enhancement and expansion. Future improvements can focus on increasing accuracy, adding advanced features, and extending real-world applicability.

One of the major areas of improvement is the support for multi-floor and complex building structures. Currently, the system primarily handles single-floor layouts, but future versions can be extended to process multi-level buildings with staircases, elevators, and interconnected floors. Additionally, the system can be enhanced to handle more complex and irregular floor plans with higher precision.

Another important direction is the integration of advanced AI models and larger datasets. Training the system with more diverse and extensive datasets can improve detection accuracy and generalization. More advanced deep learning architectures can be used to better recognize intricate architectural elements and reduce errors in detection and reconstruction.

The system can also be extended by incorporating real-time processing and optimization techniques, which would significantly reduce the time required for generating 3D models. This would make the application more efficient and suitable for real-time use cases.

Integration with Augmented Reality (AR) and Virtual Reality (VR) is another promising enhancement. This would allow users to experience immersive walkthroughs of the generated models, providing a more realistic and interactive visualization of building designs.

Furthermore, the system can include automatic interior design suggestions, where AI recommends furniture placement, color schemes, and layouts based on the room type and available space. This would make the system more useful for interior designers and homeowners.

The deployment of the system on cloud platforms can improve accessibility and scalability, allowing users to access the application from anywhere without requiring high-end hardware. Additionally, support for exporting models into formats compatible with professional tools like AutoCAD or Blender can increase its usability in industry applications.

In conclusion, with continuous advancements in AI, computer vision, and 3D technologies, PLANORA has significant potential to evolve into a comprehensive and intelligent architectural design tool capable of handling complex, real-world scenarios.

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